



BAULKHAM HILLS HIGH SCHOOL

JULY 2013
YEAR 11 TASK 2

Mathematics

General Instructions

- Working time – 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 6 to 13
- Marks may be deducted for careless or badly arranged work

Total marks – 55

Exam consists of 4 pages.

This paper consists of TWO sections.

Section 1 – Page 2

Questions 1-5 (5 marks)

- Attempt Question 1-5

Section II – Pages 3-4 (50 marks)

- Attempt questions 6 to 13

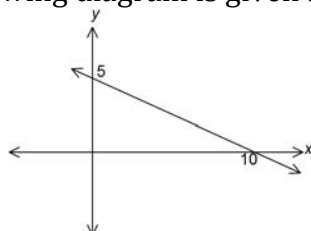
Section I - 5 marks
Attempt questions 1-5

Use the multiple choice answer sheet for question 1-5

1. Evaluate $\lim_{x \rightarrow 5} \frac{2x^2 - 11x + 5}{x^2 - 25}$

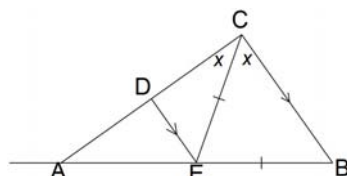
- (A) 0 (B) $\frac{9}{25}$ (C) $\frac{9}{10}$ (D) not defined

2. Equation of the line in the following diagram is given by



- (A) $y = \frac{1}{2}x + 5$ (B) $y = x + 5$
 (C) $y = -\frac{1}{2}x + 5$ (D) $y = -\frac{1}{2}x + 10$

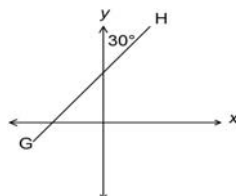
3.



In the diagram, $EB = EC$, $\angle BCE = \angle DCE$ and $BC \parallel ED$. $\triangle CDE$

- (A) is equilateral (B) is isosceles
 (C) is scalene (D) cannot be determined

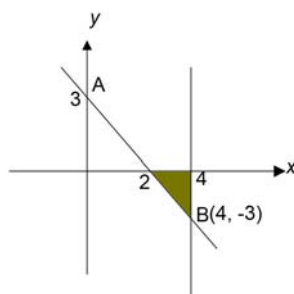
4.



The line GH makes an angle of 30° with the y -axis, as shown in the diagram. What is the gradient of GH

- (A) $\sqrt{3}$ (B) $\frac{1}{2}$
 (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

5.



The three inequalities to describe the shaded region given above are

- (A) $y \leq 0, x \geq 2, 3x - 2y + 6 \leq 0$ (B) $y \leq 0, x \leq 4, 3x - 2y - 6 \leq 0$
 (C) $y \leq 0, x \geq 2, 3x - 2y - 6 \geq 0$ (D) $y \leq 0, x \leq 4, 3x - 2y + 6 \geq 0$

Section II – Extended Response

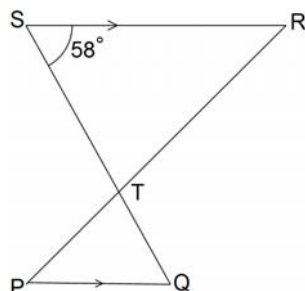
Attempt questions 6-13.

Answer each question on a SEPARATE PAGE. Clearly indicate question number.

Each piece of paper must show your name.

All necessary working should be shown in every question.

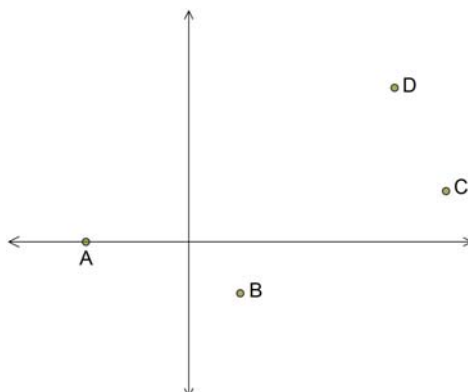
6.



Given $SR \parallel PQ$ and that SQ, RP are straight lines, and $\angle RST = 58^\circ$, $\angle PTQ = 53^\circ$, find the size of $\angle TPQ$. (without giving reason)

1

7.



Consider the quadrilateral $ABCD$ with $A(-2, 0)$, $B(1, -1)$, $C(5, 1)$ and $D(4, 3)$.
Copy the diagram with all the information given.

- (i) Find the lengths of AD and BC .
- (ii) Find the gradient of AD .
- (iii) Show that the equation of the line passing through A and D is given by $x - 2y + 2 = 0$.
- (iv) Show that $AD \parallel BC$.
- (v) What type of quadrilateral is $ABCD$?
- (vi) Calculate the perpendicular distance of B from the line AD .
- (vii) Calculate the area of $ABCD$.

3

1

2

2

1

2

2

8. Differentiate the following

(i) $y = 5x^3 + \frac{1}{x}$

(ii) $y = \frac{3x^2 + 10x}{x}$

(iii) $y = (3x + 7)(1 - x)^3$

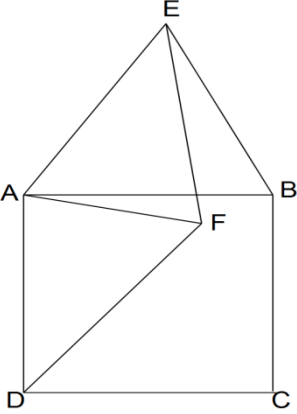
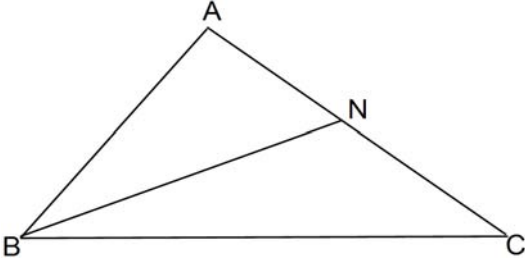
(iv) $y = \frac{5x^2 - 2}{2x + 1}$

2

2

2

2

9.	Find the derivative of the function given by $y = x^2 + 2x$ using first principles.	3
10.	Find the equation of the line which is passing through the intersection of $x - 3y + 5 = 0$ and $x + 2y = 0$ and the point $(3, 1)$.	3
11.	(i) Show that the equation of the normal to the curve $y = \frac{1}{\sqrt{x^2-3}}$ at $(2,1)$ is given by $x - 2y = 0$. (ii) Find the coordinates of the point Q where the normal meets the curve again.	2 3
12.	 In the diagram $ABCD$ is a square and AEB is an equilateral triangle. EF bisect $\angle AEB$. $\angle EAF = \angle DAF$. Copy the diagram and label the given information. (i) Show that $\angle BAF = 15^\circ$. (ii) Prove that $EF = DF$	3 4
13.	ABC is a triangle and N is point on AC. $\angle ABN = \angle CBN = \angle BCN = \theta$. $BC = 2a$, $CA = b$, $AB = c$. $BN = d$.  (i) Copy the diagram, labelling all of the given information. (ii) Explain why $CN = d$ (iii) Prove that $\angle ANB = \angle ABC$ (iv) Prove that $\triangle ABN$ is similar to $\triangle ACB$. (v) Show that $2ac = bd$ (vi) Hence show that $c^2 = b^2 - 2ac$	1 1 2 2 2 2

~ End of Exam ~

M.C

1) C 2) C 3) B 4) A

26 $\angle TPQ = 69^\circ$ — (1)

27 (i) $AD = \sqrt{(4+2)^2 + 3^2}$ — (1)
 $= \sqrt{36+9} = 4.5$ — (1)
 or $3\sqrt{5}$ — (1)

(ii) $BC = \sqrt{(5-1)^2 + 2^2}$
 $= \sqrt{20} = 2\sqrt{5}$ — (1)

(iii) $m_{AD} = \frac{3-0}{4+2} = \frac{3}{6} = \frac{1}{2}$ — (1)

(iv) $y-0 = \frac{1}{2}(x+2)$ — (1)
 $2y = x+2$
 $x-2y+2=0$ } — (1)

(v) $m_{BC} = \frac{1+1}{5-1} = \frac{2}{4} = \frac{1}{2}$ — (1)

$\therefore m_{AD} = m_{BC}$ } — (1)
 Hence $AD \parallel BC$

(vi) Trapezium — (1)

(vii) $D = \frac{|1 \times 1 - 2 \times -1 + 2|}{\sqrt{1+2^2}}$ — (1)
 $= \frac{5}{\sqrt{5}} = \sqrt{5}$ — (1)

(viii) $A_r = \frac{1}{2} \times h \times (a+b)$
 $= \frac{1}{2} \times \sqrt{5} \times (3\sqrt{5} + 2\sqrt{5})$
 $= \frac{1}{2} \times \sqrt{5} \times (5\sqrt{5})$
 $= \frac{25}{2} = 12.5 \text{ unit}^2$ with working & correct ans. (2)

Q8 (i) $y' = 15x^2 - \frac{1}{x^2}$ — (1)

(ii) $y = 3x + 10$

$y' = 3$ — (2)

(iii) $y' = 3(1-x)^3 + (3x-7) \times 3(1-x)^2 \times -1$ — (1)
 $= 3(1-x)^3 - 3(3x-7)(1-x)^2$
 $= 3(1-x)^3 [1-x - 3x+7]$
 $= 3(1-x)^3 (8-4x) = 12(1-x)^3 (2-x)$

(iv) $y' = \frac{(2x+1)(10x) - (5x^2-2)(2)}{(2x+1)^2}$ — (1)
 $= \frac{20x^2 + 10x - 10x^2 + 4}{(2x+1)^2}$
 $= \frac{10x^2 + 10x + 4}{(2x+1)^2}$

Q9 $f(x) = y = x^2 + 2x$ — (1)
 $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2(x+h) - x^2 - 2x}{h}$
 $= \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2xh + x^2 + 2x + 2h - x^2 - 2x}{h}$
 $= \lim_{h \rightarrow 0} \frac{h^2 + 2xh + 2h}{h}$
 $= \lim_{h \rightarrow 0} (h + 2x + 2)$ — (1)

$f'(x) = 2x + 2$

$$\begin{aligned} \text{Q10} \quad x-3y+5 &= 0 \\ x+2y &= 0 \\ -5y+5 &= 0 \end{aligned}$$

$$\left. \begin{aligned} y &= 1 \\ x &= -2 \end{aligned} \right\} \text{--- (1)}$$

$\therefore$ eq. of the line passing through $(-2, 1)$ & $(3, 1)$

$$m = \frac{1-1}{3+2} = 0 \text{ --- (1)}$$

$$\therefore \text{eq. is } y-1 = 0(x+2) \text{ --- (1)}$$

$$\boxed{y=1}$$

$$\text{OR} \quad \begin{aligned} x-3y+5+k(x+2y) &= 0 \text{ --- (1)} \\ 3-3k+5+k(3+2) &= 0 \\ k &= -1 \text{ --- (1)} \end{aligned}$$

$$\therefore \text{eq. is}$$

$$\left. \begin{aligned} x-3y+5-1(x+2y) &= 0 \\ x-3y+5-x-2y &= 0 \\ -5y &= -5 \\ \boxed{y=1} \end{aligned} \right\} \text{--- (1)}$$

$$\text{iii (i)} \quad y = (x^2-3)^{-1/2}$$

$$y' = -\frac{1}{2}(x^2-3)^{-3/2} \cdot 2x$$

$$= -x(x^2-3)^{-3/2} \text{ --- (1)}$$

$$m_T \text{ at } x=2$$

$$m_T = -2(4-3)^{-3/2}$$

$$= -2$$

$$m_N = \frac{1}{2} \text{ --- (1)}$$

$$\text{eq. of Normal } y-1 = \frac{1}{2}(x-2)$$

$$\Rightarrow 2y-2 = x-2$$

$$x-2y=0$$

$$\text{ii) Pt. of int. of the Normal to the curve } \left(\frac{x}{2}\right)^2 = \left(\frac{1}{\sqrt{x^2-3}}\right)^2 \text{ --- (1)}$$

$$\frac{x^2}{4} = \frac{1}{x^2-3}$$

$$x^4 - 3x^2 = 4$$

$$\text{Sub } x^2 = t$$

$$t^2 - 3t - 4 = 0 \text{ --- (1)}$$

$$(t-4)(t+1) = 0$$

$$t = 4, t = -1$$

$$x^2 = 4 \quad x^2 = -1$$

$$x = \pm 2$$

not possible

$$x = 2, y = 1$$

$$\text{or } x = -2, y = 1$$

$$\text{or } (-2, 1)$$

OR Does not touch the curve anywhere. --- (1)

Q12 In dia ABCD is a sq & AEB is an equilateral Δ . The line EF bisects $\angle AEB$ & $\angle EAF = \angle DAF$

$$\text{(i) let } \angle BAF = x$$

$$\angle EAB = 60^\circ \text{ (all sides of equilateral } \Delta \text{)}$$

$$\therefore \angle EAF = 60^\circ + x \text{ --- (1)}$$

$$\angle DAB = 90^\circ \text{ (ABCD is a sq)}$$

$$\angle DAF = 90^\circ - x \text{ --- (1)}$$

$$\left\{ \begin{aligned} 90^\circ - x &= 60^\circ + x \text{ (given } \angle DAF = \angle EAF) \\ 2x &= 30^\circ \\ x &= 15^\circ \end{aligned} \right.$$

(ii) In Δ 's DAF & EAF

$$\text{① } AD = AB \text{ (sides of a sq)}$$

$$AE = AB \text{ (sides of equilateral } \Delta \text{)}$$

$$\therefore AD = AE \text{ --- (1)}$$

$$\angle DAF = \angle EAF \text{ given}$$

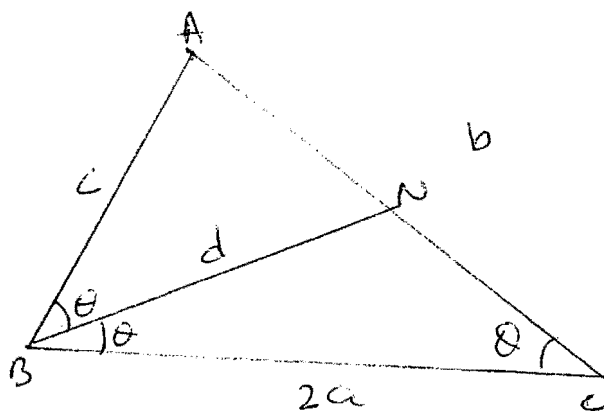
AF is common

$$\therefore \Delta AEF \cong \Delta ADF \text{ (SAS) --- (1)}$$

$$\therefore EF = DF \text{ (matching sides)}$$

$$Q \equiv \Delta s \text{ --- (1)}$$

Q13.
= (i)



- (i) $BN = CN$ (sides opp equal $\angle$ s) — (1)
 $\therefore BN = CN = d$ — (1)

- (ii) $\angle BNC = 180 - 2\alpha$ ($\angle$ sum of Δ) — (1)
 $\therefore \angle ANB = 2\alpha$ (st line $\angle$) — (1)
 Also $\angle ABC = \alpha + \alpha = 2\alpha$ — (1)
 Hence $\angle ANB = \angle ABC$ — given

- (iv) In $\triangle ABN$ & $\triangle ACB$
 $\angle BAC$ common — (1)
 $\angle ANB = \angle ABC$ (proved above)
 $\therefore \triangle ABN \sim \triangle ACB$ (all matching $\angle$ s are =) — (1)

- v) $\frac{BN}{BC} = \frac{AN}{AB} = \frac{AB}{AC}$
 (matching sides are in the same ratio in $\sim \Delta$ s) — (1)

$$\frac{d}{2a} = \frac{b-d}{c} = \frac{c}{b} \quad \text{--- (1)}$$

$$\frac{d}{2a} = \frac{c}{b}$$

$$bd = 2ac.$$

- (vi) $\frac{b-d}{c} = \frac{c}{b}$ — (1)
 $c^2 = b^2 - bd$ — (1)
 but $bd = 2ac$ (from above)
 $\therefore c^2 = b^2 - 2ac.$